

Saarlekha — Product Requirements Document (PRD)

Tagline: *Your team's reports, distilled.* **Product type:** Multi-tenant internal reporting & dashboard platform for manufacturing operations **Document version:** v1.4 (dashboard detail drill-down views, per-detail export) **Status:** Draft for development / design

1. Overview

Saarlekha is a web application that lets a company run its internal operations reporting — manpower, production, job orders, quality, and machine maintenance — from a single configurable hub. Each company defines its own report formats through an "add field" builder, and its teams view results on date-filterable dashboards.

The platform is multi-tenant: one operator (the super admin) onboards multiple companies, and each company's data is fully isolated from every other company.

2. User roles & permissions

Capability	Super Admin	Company Admin	Operations
Create / onboard companies	Yes	No	No
Create Company Admins	Yes	No	No
Create / edit / delete Operations users	Yes	Yes	No
Build & edit report formats ("add field" builder)	Yes	Yes	No
Manage manpower, customers, job-order & maintenance formats	Yes	Yes	No
Create items in Items Master	Yes	Yes	Yes (pending approval)
Approve / reject operations-created items	Yes	Yes	No
Define / edit Items Master columns	Yes	Yes	No
Assign departments to operations users (one or many)	Yes	Yes	No

Capability	Super Admin	Company Admin	Operations
Enter / modify data inside formats	Yes	Yes	Yes
Export reports	Yes	Yes	Yes
View dashboards (scoped to assigned departments)	Yes	Yes	Yes
Access admin panel / settings	Yes	Yes	No

Hard rule: Operations users can never access admin rights or admin-only screens, under any circumstance.

- **Super Admin** = the person who signs up and creates companies. Sits above all companies.
- **Company Admin** = created by a super admin; manages one company's formats and users.
- **Operations** = created by either a super admin or a company admin; data entry and export only.

3. Multi-tenancy & data isolation

- **Each company has its own dedicated database.** No shared tables across companies.
- Company A must never be able to read, query, or export Company B's data.
- Onboarding a new company provisions a new database instance for that company.
- *Implementation note:* per-company databases simplify isolation and per-client backup/export, at the cost of more databases to provision and maintain. Plan an automated provisioning step in the onboarding flow.

4. Authentication & onboarding

- **Sign-up / login via Google (OAuth).** No password storage for the primary account.
- **Company creation** captures: company name, address, logo, GST/registration number, and contact details.
- **Operations user onboarding via WhatsApp (secure pattern):**
 - Admin creates the operations user record.
 - System generates a **one-time setup/invite link** (time-limited, single-use).

- The admin shares the link using the device's **native share sheet** — sending it from their own WhatsApp (via a `wa.me` link), SMS, or email. No WhatsApp Business API is used, so there is no per-message cost to the platform.
 - The user taps the link and sets their own password — no plaintext credential is ever transmitted or stored.
 - *Rationale*: avoids exposing real passwords in chat history/backups, and keeps messaging free by relying on the admin's own WhatsApp rather than a paid API.
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5. Admin panel modules

5.1 User management

Create / edit / delete operations users (and, for super admins, company admins). Includes the WhatsApp invite-link flow above. Operations users cannot be promoted to admin.

Department assignment: when creating or editing an operations user, the admin assigns them **one or more departments**. This assignment drives what the user sees on dashboards (§6) — by default their own department, or every department assigned to them.

5.2 Manpower management

Create / edit / delete a person with the following attributes:

- Full name
- Photo
- Contact details
- **Aadhaar number** — **always stored and displayed masked** (only the last 4 digits visible), per Indian law (UIDAI masking norms / DPDP Act, 2023). The full number is never stored in readable form. Confirm the final approach with a qualified compliance advisor.
- Blood group
- Emergency contact number
- Role
- Department assignment (dropdown)

5.3 Report builder (core module — highest complexity)

- **"Add field" builder model**. The admin builds a report format by adding fields one at a time: an "+ Add field" action opens a form (a bottom sheet on mobile) where the admin defines that field — its name, data type, and unit of reporting — then adds it to the format. Repeat for as many fields as needed; fields can be grouped into sections/tabs.

- The format is shown as a **single-column list of field cards**; fields can be reordered, edited, and deleted. (This is the mobile-native equivalent of a drag-and-drop builder — same capability, every action is a tap.)
- Each defined field carries a **data type** (so the system knows how to store, validate, and aggregate it — e.g. number fields can feed production figures and targets).
- Define report formats, SKUs, and units of reporting.
- **Edit and delete** existing fields and formats from the admin panel.
- **Format versioning — preserve old data (confirmed)**: when a field is edited or deleted after data already exists, existing records keep their original fields and stay fully viewable and exportable; the change only affects *new* entries going forward. No historical data is ever lost.

5.4 Job order builder

- Define parameters for job-order management reports.
- Create units of material.
- Select customer from a dropdown (pulls from Customer Master, §5.5).

5.5 Customer master

Create / edit / delete customers. Feeds the customer dropdown in the Job Order builder.

5.6 Machine maintenance report builder

Define formats/parameters for machine maintenance reporting (analogous to the report builder, scoped to maintenance).

5.7 Items master (with approval workflow)

A master list of items the company uses, with **customizable columns** and an **approval gate** for operations-created entries.

Customizable columns: like the report builder, the item master's columns are not fixed. An admin defines the fields each item carries (e.g. Item name, Item code/SKU, Unit, Category, Rate, plus any custom fields) using the same "add field" model — name, data type, unit — and can edit or delete columns later. Column edits/deletes preserve existing item data (same versioning rule as §5.3).

Who can do what:

- **Admin / Super admin**: create, edit, delete items directly; define and manage the item-master columns; review and authorize items submitted by operations.
- **Operations**: can **create** a new item, but the item is saved as "**Pending approval**" and is *not* usable elsewhere in the app until authorized. Operations cannot edit the column structure, and cannot approve items (including their own).

Approval flow:

1. Operations user fills in the item fields and submits → item status = **Pending**.
2. The item appears in an admin **approval queue** (with who submitted it and when).
3. Admin/super admin **approves** (status = Active, item becomes selectable in dropdowns elsewhere) or **rejects** (with an optional reason sent back to the submitter).
4. Every create / approve / reject / edit / delete is written to the audit log (§8).

Status values: Pending, Active, Rejected (and optionally Inactive for retiring an item without deleting its history).

6. Dashboards & reports

Visible to **both admin and operations**. Every report supports a **date / date-range selector** for the reporting period.

Report	Contents
Quality reports	As defined in the report builder
Job order reports	Built from job-order formats; linked to customers & materials
Production reports	Machine-wise and operator-wise, including efficiency (see §7)
Manpower summary	Headcount, departments, roles from manpower module
Machine maintenance reports	Built from the maintenance format

Dashboard scope (access rule, not just a UI filter): an operations user sees data for **their own department only**, unless an admin has assigned them **multiple departments**, in which case they see all departments assigned to them. Admins and super admins see everything in the company.

Two levels — summary and detail:

- The dashboard shows **summary cards** (KPIs + a card/section per report type).
- **Tapping a report card opens its detail view.** Dedicated detail screens exist for **Quality, Job Order, Production, and Machine Maintenance**, each showing the full records/breakdown for the selected date range (e.g. production detail by machine and operator with efficiency; maintenance detail per machine).
- Each detail view respects the same date-range and department filters carried in from the dashboard.

- **Each detail view is independently exportable** (Excel, PDF, CSV, TXT) via the same download-then-share flow as §9 — so a user can export a single report (e.g. just Production) without exporting the whole dashboard.
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7. Efficiency calculation

For the selected reporting period, **operator-wise and machine-wise efficiency** is:

$$\text{Efficiency} = (\text{total production figure for the period}) \div (\text{sum of daily targets for the same period})$$

Tips / data rules:

- **Target is always defined per day.** The period total is the sum of those daily targets.
 - **Production figure is always entered per day.** The period total is the sum of those daily entries.
 - Because the division happens once on the period totals (not per day), there is no per-day divide-by-zero issue.
 - Edge case: if the **sum of targets for the period is 0**, display efficiency as N/A rather than dividing by zero.
 - Decide whether to **cap at 100%** or allow values above 100% when production beats target (recommend: allow above 100%, since it reflects real over-performance).
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8. Audit log

- Log every **create / edit / delete** across people, customers, formats, and data entries.
 - Capture: who (user), what (entity + change), and when (timestamp).
 - Surfaced to admins; immutable (append-only).
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9. Exports

- **Formats:** Excel, PDF, CSV, TXT.
- **Delivery:** the file is downloaded to the device, then shared via the **native share sheet** (the user sends it through their own WhatsApp, email, or any app). Note: a `wa.me` link cannot attach a file, so exports are always download-then-share rather than auto-sent; this keeps messaging free (no WhatsApp Business API cost).

- *Privacy note:* exports may contain personal data (e.g., masked Aadhaar, contact details). Gate exports to authorized roles, log every export in the audit trail, and never include the full Aadhaar number in any export.
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10. Data entities (high-level)

- **Company** (name, address, profile)
 - **User** (role: super admin / admin / operations, auth, company link, **assigned department(s)** for operations users)
 - **Person / Manpower** (name, photo, contact, masked Aadhaar, blood group, emergency contact, role, department)
 - **Department**
 - **Customer**
 - **Item** (master record with admin-defined custom columns; status: Pending / Active / Rejected / Inactive; submitted-by, approved-by, timestamps)
 - **Material unit**
 - **Job order**
 - **Report format** (with user-defined fields + data types, SKUs, units; versioned to preserve historical data) + **Report entries**
 - **Machine + Maintenance record**
 - **Production record** (daily production figure, daily target, operator, machine)
 - **Audit log entry**
 - **Export record**
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11. Compliance & security checklist

- Aadhaar always masked (last 4 digits only); full number never stored in readable form; encrypt sensitive fields at rest.
- Confirm Aadhaar handling with a qualified compliance/legal advisor (UIDAI rules + DPDP Act, 2023).
- Never transmit or store plaintext passwords; use one-time setup links.
- Per-company database isolation enforced at the data-access layer.
- Role-based access control with operations users hard-blocked from admin functions.
- Audit logging on all sensitive operations.
- Export gating + logging; no full Aadhaar in exports.

12. Resolved decisions

1. **Company profile fields:** name, address, logo, GST/registration number, contact details. ✓
2. **Operations dashboard scope:** own department only, unless an admin assigns multiple departments (then all assigned). Admin functionality to assign one-or-many departments included. ✓
3. **Report-format versioning:** preserve old data — edits/deletes affect new entries only; historical records keep their original fields and stay viewable/exportable. ✓
4. **Efficiency:** total production for the period ÷ sum of daily targets for the period. Target and production are always entered per day. ✓
5. **Field builder:** admin adds unlimited fields (tabs) and defines each (name, data type, unit). ✓
6. **Aadhaar:** always masked. ✓

Remaining to confirm

- Cap efficiency at 100% or allow above 100% (recommend: allow above).
- Confirm Aadhaar masking approach with a qualified compliance advisor.

This PRD captures the agreed scope and decisions to date. It is a living document — update versioning as decisions in §12 are resolved.